Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Butel, Janet S. Ph.D.		
Project Title:	Viral Infections and Mucosal Immunity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	25
Comments:	Address updated 9/2008		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	03/31/2004
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Conner, Margaret (Baylor College of Medicine) Lednicky, John (Baylor College of Medicine) Ling, Paul (Baylor College of Medicine) Pierson, Duane (NASA JSC) Shearer, William (Baylor College of Medicine) Sonnenfeld, Gerald (Binghamton University, State University of New York)		
Grant/Contract No.:	NCC 9-58-IIH00204		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space flight has been found to affect immune responses, although the underlying mechanisms have not been established. Alterations in a normal immune response could have a major impact on the host's ability to control infections. An important question being explored is whether infectious diseases will pose an unacceptable medical risk to the success of long-duration space journeys. All humans are infected for life with latent and persistent viruses, and it is well-known that suppression of the immune system allows latent viruses to reactivate and multiply, which may cause disease in the person undergoing reactivation or in contacts to whom the virus is transmitted. The general hypothesis being addressed is that conditions of long-duration space flight, including stress, isolation, containment, microbial contamination, and solar radiation, will alter human immune responses, leading to reactivation of latent viruses, increased viral infections and viral disease, and possible development of malignancies. We expect that immune changes will include altered mucosal immunity, an important host defense against microbial infections. We are focusing on reactivation and shedding of human herpesvirus EBV and human polyomaviruses, agents known to establish persistent infections and to undergo reactivation and cause disease, including cancer, when the host immune system is compromised. Animal models are being used to study radiation effects on host responses to infections. The original aims of the project were: (1) To determine the effect of space flight conditions on virus reactivation, infection, and replication using ground-based models that simulate aspects of space flight; (2) To determine the effect of irradiation on the immune system, susceptibility to virus infection, and development of virus-induced cancers in murine animal models; and (3) To characterize global changes in mucosal immune responses under simulated space flight conditions in the hind limb unloading (suspended mouse) model. Key findings of the project this year were from the mouse polyoma virus space radiation model we developed. Preliminary studies suggest that combined effects of radiation and virus infection on the immune system of experimental animals lead to immunosuppression and latent virus reactivations. Because of the observed effects upon immunocompetence and viral replication, it is likely that future studies will be able to offer surrogate marker tests for virus infection and immunocompromise and new approaches to prevention and treatment of latent virus reactivation and lymphoid malignancies that develop as a consequence.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight-induced alterations in the immune system, if serious enough, would have marked adverse effects on host control of microbial infections. Consequences of altered host immunity could result in virus reactivation, replication, and disease development, including cancer. As all humans are infected for life with latent or persistent viruses, those infections will be uninvited travelers on all space missions, and it is prudent to understand the implications of their presence. This project is addressing immune system deficits and reactivations of herpesviruses and polyomaviruses, two important classes of viruses that target different tissues in the body and serve as sentinels of the effects of space flight conditions on latent viral infections in general. We have studied ground-based human models of space flight — two that model confinement and isolation [wintering-over in Antarctica and a closed-chamber study (Russian)] and one that models different degrees of immune damage (HIV-infected individuals). Virus reactivation and shedding results were interpreted by comparison to the baseline established from a year-long longitudinal study of healthy volunteers. It is noteworthy that HIV-infected individuals are susceptible to a number of opportunistic infections and cancer, including those linked to EBV and polyomavirus. These subjects are a model to predict long-range health complications in slightly to moderately immune compromised persons. The data show the crucial role that host immune function plays in regulation of latent viral infections. Even modest depressions in immune function correlate with virus reactivation and shedding (both EBV and JCV). It is evident that viral contamination within a spacecraft is an issue to be reckoned with and considered. These types of data will guide decision-making regarding countermeasure development, including vaccines to boost antiviral immunity, the use of immune modulators to stimulate components of the immune system to keep viral infections in check, and antiviral pharmaceuticals to treat reactivated viruses. Precautions might need to be designed and implemented to inactivate and minimize the spread of excreted viruses in the spacecraft environment. The evidence for viral reactivations highlights the need to understand synergistic effects of radiation and space flight conditions that may result in more deleterious effects on immune function and control of microbial infections and disease. Of concern is that the agents we have shown to be reactivated have been linked with human cancers, and cancer development is known to be exacerbated when host immune functions are compromised. We have established a murine model using polyoma virus to analyze the effect of radiation on immune parameters and susceptibility to virus infection. We have shown that radiation can cause reactivation of latent viral infections and depress host immune function. The knowledge gained from studies of virus reactivations in these test models will be applicable to earth-bound individuals at risk of suffering similar virus reactivations and serious, sometimes life-threatening, consequences due to immunosuppression following organ transplantation or cancer chemotherapy and during pregnancy, old age, and AIDS.
Task Progress:	Progress was made on the development and validation of the mouse polyoma virus space radiation model. We determined optimal doses of viral inocula, which organs to analyze, optimal times of harvest of specimens to analyze, kinetics of viral clearance/entry into latency, and tissue processing methodology; we developed a PCR assay for the virus. We confirmed that proton irradiation and gamma irradiation had similar effects on the murine immune response. We showed that whole-body gamma irradiation delayed clearance of primary virus infection and reactivated latent viral infections. Radiation reduced spleen cell counts and lymphocyte proliferation potential. It blunted gamma interferon production (a cytokine important in controlling viral infections and cancer). Our findings suggest that combined effects of radiation and virus infection on the immune system lead to immunosuppression and latent virus reactivations.
Bibliography Type:	Description: (Last Updated: 04/25/2011)
Articles in Peer-reviewed Journals	Lednicky JA, Halvorson SJ, Butel JS. "PCR detection and DNA sequence analysis of the regulatory region of lymphotropic papovavirus in peripheral blood mononuclear cells of an immunocompromised rhesus macaque." J Clin Microbiol. 2002 Mar;40(3):1056-9. PMID: 11880438 , Jan-2003

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Awards	Butel JS. "Janet S. Butel: Distinguished Alumnus of the Graduate School of Biomedical Sciences Award, Baylor College of Medicine, 2002. January 2003." Jan-2003
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Presentation	Butel, J. S., J. A. Lednicky, R. A. Vilchez, F. Visnegarwala, D. E. Lewis, C. A. Kozinetz, and W. A. Keitel. "Polyomavirus JCV reactivation and shedding in healthy and immunocompromised hosts: implications for space travel. " Jan-2003
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Presentation	Reuben, J. M., J. S. Butel, B. N. Lee, S. Zhang, D. Y. Shen, and W. T. Shearer "The effect of gamma irradiation on the immune responses of mice with polyoma virus infection. " Apr-2003
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